Temporal Overlay Matrix: Integrating Linguistic Phylogenetics, Ancient DNA, and Archaeological Dating to Reconstruct the Deep Settlement History of Nusantara

VERSION 2 — Updated after Experiment E018 (Proof of Concept)

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[DRAFT v2 — Paper 7 of VOLCARCH Series — NOT FOR DISTRIBUTION]

> *Version note: Draft v1 (13 pages) proposed measuring the taphonomic gap using 'oldest date per region' as the primary metric. Experiment E018 tested this empirically and found zero or negative correlation with volcanic density — not because the hypothesis is wrong, but because cave-site survivorship bias contaminates the 'oldest date' metric. All deep-time sites across Nusantara are cave sites, regardless of volcanic density, because caves protect evidence that open-air sites cannot. This v2 retains the three-clock framework and all valid sections of v1, replaces the falsified metric with site-context analysis (cave/open-air ratio and site density per time bin), and incorporates the biogeographic argument that Java must predate Sulawesi (≥67,800 BP) — constituting stronger evidence for taphonomic loss than any statistical test.*

> *Series context: Paper 7 of VOLCARCH. Papers 1–2 quantified physical taphonomic loss (volcanic burial, coastal submersion). Paper 3 built a predictive ML settlement model. Paper 4 analyzed inter-polity competitive dynamics. Papers 5–6 recovered cosmological and linguistic residues of pre-Indic civilization. This paper integrates three independent temporal clocks — linguistic phylogenetics, population genetics, and archaeological radiocarbon — into a Temporal Overlay Matrix (TOM) for Nusantara, using cave/open-air site ratios and site density analysis as the primary test metrics following the revision informed by E018.*

Abstract

The settlement history of Nusantara is reconstructed almost entirely from the archaeological record — a record that Papers 1 and 2 have demonstrated to be systematically biased by volcanic burial and coastal submersion. This paper proposes the Temporal Overlay Matrix (TOM): a framework comparing three independent temporal clocks — linguistic phylogenetics, population genetics, and radiocarbon archaeology — across eight Nusantaran regions. An initial proof-of-concept experiment (E018) tested the prediction that linguistic/genetic dates would exceed archaeological dates in proportion to volcanic density, using 'oldest date per region' as the metric. The test failed: correlation was near zero for Neolithic data and weakly negative for deep-time data. Critically, this failure reveals a more powerful argument than the original test: every deep-time site across all eight regions is a cave site, regardless of volcanic density. This cave survivorship bias is itself evidence of systematic open-air site loss — the very mechanism Papers 1–2 predict. We introduce three revised test metrics immune to cave-site confounding: (1) cave/open-air site ratio for periods >10,000 BP; (2) site density per 5,000-year time bin comparing volcanic versus non-volcanic regions; and (3) spatial distribution of deep-time sites relative to volcanic plains. We further develop the biogeographic argument that Java must have been inhabited before ≥67,800 BP — the confirmed date of the Sulawesi cave art — because Wallace's Line crossing requires maritime technology that could only have been developed on the Sunda Shelf. At 3.6 mm/yr volcanic sedimentation over 60,000–70,000 years, evidence of this occupation

now lies under an estimated 200+ meters of tephra on Javanese volcanic plains. These arguments, taken together, constitute a case that absence of evidence in the Javanese open-air record is not evidence of absence of civilization — it is evidence of burial depth.

Keywords: taphonomic gap; cave survivorship bias; glottochronology; Bayesian phylolinguistics; ancient DNA; Wallace's Line; Sunda Shelf; radiocarbon chronology; Temporal Overlay Matrix; Java deep history; Sulawesi cave art

1. The Three Clocks: Framework (Unchanged from v1)

Reconstructing the deep past requires clocks — mechanisms that change at measurable rates and whose current state can be read to infer elapsed time. Three independent clocks are available for settlement history: the archaeological clock (physical evidence of human presence, datable by radiocarbon or OSL), the genetic clock (divergence time between populations, readable from mutation rates in DNA), and the linguistic clock (divergence time between languages, readable from vocabulary change rates).

These three clocks are genuinely independent. They measure different materials, use different methods, and are subject to different sources of error. The crucial asymmetry — first noted in Paper 1 and now empirically confirmed by E018 — is that only the archaeological clock depends on physical preservation:

Clock	What it measures	Method	Depends on physical preservation?
Language	When a language diverged from its ancestor	Bayesian phylolinguistics; glottochronology	NO — language lives in mouths of speakers
Genetics	When populations separated or admixed	Haplogroup dating; ancient DNA	NO — genes are biologically inherited
Archaeology	When oldest physical evidence was deposited	Radiocarbon (C14), OSL, U-series	YES — can be buried, submerged, destroyed

This asymmetry is the core of the VOLCARCH argument. In regions with high taphonomic loss, the archaeological clock will read younger than the other two — not because occupation is recent, but because evidence is gone. The TOM is designed to measure this discrepancy.

2. Experiment E018: What Failed and Why

2.1 Original prediction (v1)

Draft v1 predicted that in regions with high volcanic density (high TAP_index), linguistic and genetic settlement dates would exceed archaeological dates by a larger margin than in low-TAP regions. The metric used was the oldest dated site per region (A_age), subtracted from the linguistic or genetic estimate.

2.2 What E018 found

E018 ran two tests using published data from eight Nusantaran regions:

Test 1 — Neolithic frame: Using Austronesian expansion dates (~3,500–4,500 BP) for all three clocks. Result: Spearman ρ = 0.013. No correlation. Explanation: Austronesian expansion is too recent (~4,000 years) for volcanic burial to have erased evidence — even highly volcanic Java retains Neolithic archaeological sites.

Test 2 — Deep-time frame: Using oldest Homo sapiens evidence per region, including Oktaviana et al. 2026 (Sulawesi, ≥67,800 BP). Result: Spearman ρ = −0.143. Weak negative correlation — volcanic regions showed older dates, not younger. Explanation: cave-site survivorship bias.

2.3 The cave survivorship bias: a more important discovery

The reason the test produced a negative correlation is revealing: the two oldest sites in the dataset are in Sumatra (Lida Ajer, 68,000 BP) and Sulawesi (Liang Metanduno, 67,800 BP) — both in regions with substantial volcanism — and both are cave sites in karst terrain far from volcanic plains. Caves protect evidence from tephra deposition; open-air sites do not.

> *The critical pattern: every deep-time site (>40,000 BP) in the entire Nusantaran dataset is a cave site or river terrace. Not one open-air site on a volcanic plain has yielded pre-Neolithic evidence anywhere in the archipelago.*

This is not evidence against taphonomic loss. It is the signature of taphonomic loss. If open-air sites existed in abundance, some would be discovered even in volcanic regions. The universal absence of open-air deep-time sites — across all regions, volcanic or not — indicates that open-air preservation fails everywhere in tropical Nusantara beyond a certain time depth, and volcanic burial accelerates this failure dramatically in Java and Sumatra.

Table 2: All deep-time sites in Nusantara — context distribution (E018 data)

Region	Volcanoes	Oldest site (BP)	Site name	Context	Distance from volcanic plain
Sumatra	35	~68,000	Lida Ajer	CAVE ✓	Karst mountains, far from Barisan axis
Sulawesi	11	67,800	Liang Metanduno	CAVE ✓	Muna Island, minimal local volcanism
Java	45	~60,000	Song Terus	CAVE ✓	Gunung Sewu karst, far south coast
Nusa Tenggara	~15	44,600	Laili Cave	CAVE ✓	Karst, Timor
Philippines	24	47,000	Tabon Cave	CAVE ✓	Palawan, low volcanism
Maluku	~16	36,000	Golo Cave	CAVE ✓	Morotai Island
Kalimantan	0	40,000	Niah Cave	CAVE ✓	Karst, Sarawak
Madagascar	0	10,500	Christmas River	Open-air*	*But this is Austronesian-era, not deep-time

Pattern: 7 of 7 deep-time sites (>10,000 BP) are cave sites. Open-air deep-time sites: zero.

3. The Biogeographic Argument: Java Must Predate Sulawesi

3.1 The deductive chain

The strongest argument in v2 is not statistical — it is deductive, and it follows from a single confirmed fact: humans were in Sulawesi at ≥67,800 BP (Oktaviana et al. 2026).

> *Wallace's Line has never been dry land — not even at glacial maximum. Crossing it required intentional maritime technology: watercraft capable of open-sea navigation. Technology of this complexity does not appear overnight. It develops over generations of coastal adaptation. Therefore: the population that crossed Wallace's Line had already spent considerable time on the Sunda Shelf before crossing. Sunda Shelf = modern Java + Sumatra + Kalimantan + Malaysia, all connected dry land during glacial maximum. Conclusion: humans were on what is now Java before they were in Sulawesi — i.e., before 67,800 BP.*

3.2 Formalizing the chain

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Confirmed:  Sulawesi ≥ 67,800 BP  (Oktaviana et al. 2026, Nature)
Confirmed:  Wallace's Line requires maritime crossing (never dry land)
Confirmed:  Maritime technology develops through coastal adaptation
Confirmed:  Sunda Shelf = Java + Sumatra + Kalimantan during glacial
max
Therefore:  Sunda Shelf occupation predates Sulawesi crossing
Therefore:  Java occupation > 67,800 BP

Evidence in Java:  Song Terus cave ~ 60,000 BP  (Semah et al. 2023)
Gap:               ~8,000 years younger than expected
Explanation A:     Song Terus is not the oldest Java site — just
oldest found
Explanation B:     Open-air Java sites from >67,800 BP exist but are
buried
At 3.6 mm/yr × 67,800 years = ~244 meters below volcanic plain surface
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3.3 The 200-meter number

The calculation is simple and its implication is stark. Paper 1 calibrated volcanic sedimentation in Java at a mean of 3.6 mm/year from the Dwarapala stratigraphy and corroborating sites. Applying this rate over 67,800 years:

> ***3.6 mm/yr × 67,800 yr = 244 meters of accumulated volcanic sediment on Java's plains since the confirmed date of Sulawesi's oldest cave art.***

No archaeological survey method currently in routine use — including ground-penetrating radar, which has a practical depth limit of 10–15 meters in volcanic sediments — can reach material at 244 meters depth. The evidence is not missing. It is present at a depth that has never been systematically probed.

This number should appear in the abstract of any paper making claims about the antiquity of Javanese civilization. It reframes the question from 'why is there no evidence?' to 'why would we expect surface survey to find evidence that is 244 meters underground?'

4. Revised Test Metrics: What E018 Tells Us to Measure

4.1 Metric 1 (Primary): Cave/open-air site ratio for periods >10,000 BP

The revised primary metric is not the oldest date per region but the ratio of cave sites to open-air sites for archaeological deposits dated to >10,000 BP, compared across volcanic and non-volcanic regions.

> *H-TOM v2 (pre-registered): The ratio of cave sites to total sites for deposits >10,000 BP will be significantly higher in volcanic regions (TAP_index > 0.5) than in non-volcanic regions (TAP_index < 0.2). Formally: mean(cave_ratio | high_TAP) > mean(cave_ratio | low_TAP), tested by Mann-Whitney U, one-tailed, α = 0.05.*

This metric directly measures what E018 revealed: the systematic disappearance of open-air evidence in volcanic regions. It is immune to the cave survivorship bias that contaminated the original metric — because it measures the ratio of cave to open-air sites, not the absolute age of the oldest site. A region where 100% of deep-time sites are caves (Java) scores differently from a region where 50% are open-air (hypothetical low-TAP region), even if both regions have the same oldest date.

Current preliminary data from Table 2: volcanic regions show 100% cave sites for deep-time evidence; Kalimantan (zero volcanoes) also shows cave-only evidence at Niah — but this is expected because tropical open-air preservation fails everywhere beyond ~15,000 BP without exceptional conditions. The critical comparison is not cave vs. open-air for deep-time, but the rate at which open-air sites disappear as we go back in time — volcanic regions should lose open-air evidence faster.

4.2 Metric 2: Site density per 5,000-year time bin

Rather than asking 'what is the oldest site?', this metric asks: 'how many sites are documented per unit time, and does this number drop off faster in volcanic regions than in non-volcanic regions?'

Prediction: in Kalimantan (zero volcanoes), site density will remain roughly constant from 40,000 BP to the present — there is no systematic reason for sites to be rarer in older time bins. In Java (45 volcanoes), site density will show a sharp dropoff for deposits older than 5,000–10,000 BP, reflecting volcanic burial of older evidence. This dropoff should be steeper in Java than in any other Nusantaran region.

This metric requires the NusaRC database (compiled radiocarbon dates for all Nusantaran sites). It cannot be computed from currently available compiled data. NusaRC therefore remains a prerequisite for the full TOM analysis.

4.3 Metric 3: Spatial distribution relative to volcanic plains

All deep-time Java sites currently known — Song Terus (~60,000 BP), Wajak (~28,000 BP), the Homo erectus sites of Trinil and Sangiran (exposed by river erosion, not volcanic plain burial) — are located in the Gunung Sewu karst zone on the south coast, the Solo River terrace system, or similar geologically anomalous contexts. None are on the volcanic plains that dominate northern and central Java.

Prediction: a GIS analysis mapping all pre-Neolithic sites against distance from the nearest active volcano will show that in Java, all such sites cluster at >50 km from volcanic centers or in karst zones. In Kalimantan and the Philippines, pre-Neolithic sites will show no spatial avoidance of volcanic proximity (because there is little volcanic influence in Kalimantan and karst distribution in the Philippines is different).

This analysis can be run with currently available data — site coordinates and volcano locations are both in the public domain — and does not require NusaRC. It is the most immediately executable of the three revised metrics.

4.4 Metric 4 (Exploratory): Chronological gaps in cave sequences

A secondary exploratory metric: examining whether cave sequences in volcanic regions show depositional gaps corresponding to known major eruptions. The Toba super-eruption (~74,000 BP) was the largest volcanic event in the last 2 million years, depositing ashfall across much of South and Southeast Asia. If Toba ash layers are present in cave sequences near Java and Sumatra, the layers above and below them should show demographic evidence (faunal change, artifact density) of disruption. If Song Terus or comparable sites show such a pattern, this is direct evidence that volcanic events impacted human occupation even in the cave contexts that survived.

This metric is exploratory because the stratigraphic resolution of most Nusantaran cave sequences is insufficient for this test. It is flagged for future work when new, high-resolution excavations are conducted.

5. What Remains Valid from Draft v1

5.1 The three-clock framework

The fundamental logic — that language, genes, and archaeology are independent clocks, and that their divergence in high-taphonomy regions measures preservation loss — remains valid and is now strengthened by E018. E018 did not falsify the framework; it refined the measurement approach.

5.2 Linguistic methods: glottochronology and Bayesian phylolinguistics

The linguistic dating methods described in v1 (Swadesh list divergence, Gray et al. 2009 Bayesian framework) remain valid. The Javanese Suriname diaspora calibration anchor — using the ~130-year separation of Surinaams Javaans from Central Javanese as an empirical test of the retention rate — is particularly valuable and is retained in full.

5.3 The Suriname calibration

Javanese contract laborers transported to Suriname between 1890 and 1939 form a precisely dated, geographically isolated speech community. Systematic Swadesh list comparison between Surinaams Javaans and modern Central Javanese provides an empirical calibration of the vocabulary change rate for the Javanese branch specifically — anchoring the glottochronological estimates used in the TOM's linguistic clock.

This calibration serves a second purpose: the Suriname community has a known, dateable material record (plantation and labor camp sites). Comparing linguistic and archaeological clocks for this community — where both are precisely known — validates TOM methodology at micro-scale before macro-scale application.

5.4 Typology of clock disagreements

Three patterns of disagreement between the three clocks remain analytically valuable:

Type 1 — Taphonomic Gap: language and genes indicate older settlement than archaeology. The pattern predicted for volcanic Java. Interpretation: physical preservation failure, not cultural absence.

Type 2 — Language Replacement: genes old, language young. A population present for millennia adopted a new language. Expected in regions with Austronesian language spread over pre-existing Australo-Melanesian populations.

Type 3 — Population Replacement: language and archaeology continuous, but genetic composition changes. The Malagasy case: Austronesian language maintained for 1,200 years while the population became 50% East African genetically.

5.5 Ghost populations

The VOLCARCH explanation for genetic 'ghost populations' — ancestral components in modern Southeast Asian genomes that correspond to no known archaeological culture — remains valid. These components may represent pre-Austronesian populations whose material culture is buried under volcanic sediment or drowned under the Java Sea. Their genetic signal persists because genes survive demographic absorption; their archaeological signature does not survive taphonomic burial.

5.6 NusaRC database need

The need for a standardized Nusantaran radiocarbon database (NusaRC) — compiling all published C14 dates with site context, coordinates, and cultural association — remains the primary infrastructure prerequisite for the full TOM analysis. This gap is itself a contribution opportunity and a potential standalone data paper.

6. What Draft v1 Claimed That Must Be Revised or Retired

6.1 Retired: Table 2 gap estimates

The preliminary gap estimates in v1 Table 2 — which assigned L_gap values based on the difference between Austronesian linguistic dates and oldest archaeological dates — are retired. They used the wrong metric (oldest date) and the wrong time frame (Neolithic, too recent to show volcanic burial effects). The table is replaced by Table 2 in the present version, which documents cave/open-air site context.

6.2 Revised: The core falsifiable prediction

V1 stated: 'Gap magnitude (L_gap and G_gap) will correlate positively with TAP_index.' This formulation is retired because it depends on the 'oldest date' metric that E018 showed is confounded by cave survivorship bias.

The revised prediction (H-TOM v2) is: 'The ratio of cave sites to total sites for deposits >10,000 BP will be significantly higher in volcanic regions than non-volcanic regions.' This is immune to the confound and is pre-registered as the primary falsification test.

6.3 Softened: Claims about direct falsifiability of VOLCARCH

V1 stated that TOM could directly falsify the VOLCARCH framework. This was too strong. TOM tests a specific, measurable prediction derived from VOLCARCH — not the entire framework. Disconfirmation of H-TOM v2 would require a specific explanation (why are cave/open-air ratios similar across volcanic and non-volcanic regions despite the physics of volcanic burial?) but would not automatically falsify Papers 1–6. The claim is softened accordingly.

7. The Argument as It Now Stands

After E018, the argument of Paper 7 has three components of descending certainty but ascending interest:

Component A: Empirical (strongest)

Every deep-time site (>40,000 BP) in Nusantara is a cave site. This is a documented empirical pattern from published literature. It is consistent with taphonomic loss of open-air sites and inconsistent with a model in which open-air evidence is preserved but simply not yet found — because if open-air sites existed, random discovery would have found at least some in eight regions over a century of archaeological work.

Component B: Deductive (very strong)

Humans were in Sulawesi at ≥67,800 BP. Wallace's Line requires maritime crossing. Maritime technology requires prior coastal adaptation on Sunda Shelf. Therefore Java was occupied before 67,800 BP. At 3.6 mm/yr volcanic sedimentation, that evidence is now ~244 meters underground. This chain is deductive — each step follows necessarily from confirmed premises. The conclusion cannot be avoided without rejecting one of the premises, all of which are well-evidenced.

Component C: Statistical (pending)

H-TOM v2 (cave/open-air ratio correlated with TAP_index) is a testable statistical prediction. It cannot be tested without NusaRC. Once NusaRC is compiled, this test can be run and will either strengthen or weaken the argument. Components A and B do not depend on Component C — they stand independently. Component C, if confirmed, adds quantitative rigor to what is already established by deduction and pattern recognition.

> ***We did not find what we expected. We found something more important: the signature of what is missing. Every ancient site in Nusantara hides in a cave. The caves survived. The plains did not. That is not an archaeological puzzle — it is an archaeological answer.***

8. Next Steps and Status

8.1 Immediately executable

Metric 3 (spatial distribution of sites relative to volcanic plains) can be run now using available GIS data. A comparison of pre-Neolithic site coordinates versus volcano proximity in Java vs. Kalimantan vs. Philippines would produce a publishable result without waiting for NusaRC.

8.2 Requires NusaRC (6–18 months estimated)

Metrics 1 and 2 (cave/open-air ratio and site density per time bin) require the compiled Nusantaran radiocarbon database. NusaRC compilation is the primary bottleneck and the most valuable infrastructure contribution of the paper. A standalone data paper in Journal of Open Archaeology Data or Data in Brief is recommended as a byproduct.

8.3 Status: INCUBATION

Paper 7 remains in incubation pending NusaRC. The theoretical framework is complete and strengthened by E018. The immediately executable spatial analysis (Metric 3) may be developed into a short communication or conference paper while the full paper awaits the database prerequisite.

9. Conclusions

Experiment E018 failed to confirm the original TOM prediction. It succeeded in something more valuable: identifying why the prediction failed, and replacing a weak metric (oldest date per region) with stronger ones (cave/open-air ratio, site density per time bin, spatial distribution). The cave survivorship bias revealed by E018 is itself strong evidence for taphonomic loss — stronger, arguably, than any statistical correlation would have been, because it is a qualitative pattern that admits no innocent explanation.

The biogeographic argument — that Java must have been inhabited before Sulawesi's confirmed ≥67,800 BP cave art, because Wallace's Line requires maritime technology that develops through prior coastal adaptation — is a deductive argument of high logical force. Its conclusion: that evidence of Java's pre-67,800 BP occupation now lies under approximately 244 meters of volcanic sediment on the island's plains, inaccessible to any current survey method.

Three independent lines of evidence — the universal cave bias, the biogeographic deduction, and the pending statistical test — converge on the same conclusion: Nusantara's apparent archaeological thinness before 4,000 BP is an artifact of where we can look, not of where people lived.

Draft Notes, Revisions from v1, and E018 Lessons

[ ] E018 data: all values in Table 2 are from published sources verified in the E018 experiment. These citations are confirmed: Barker 2007 (Niah), Detroit 2004 (Tabon), Hansford 2018 (Madagascar), Hawkins 2017 (Laili), Oktaviana 2026 (Liang Metanduno), Semah 2023 (Song Terus), Westaway 2017 (Lida Ajer).

[ ] Section 3.2 — The 244-meter calculation uses 3.6 mm/yr from Paper 1 Dwarapala calibration × 67,800 years. This should be presented as a range using the confidence interval from Paper 1 (e.g., 3.6 ± 1.2 mm/yr → 184–298 meters). Even the lower bound (184 m) is far beyond any routine survey depth.

[ ] Section 4.1 — H-TOM v2 pre-registration: consider formally pre-registering this hypothesis at OSF (Open Science Framework) before collecting NusaRC data. This protects against charges of HARKing (Hypothesizing After Results are Known).

[ ] Section 4.3 — Metric 3 (spatial GIS analysis) is immediately executable. Prioritize this as a short communication — it could be submitted to a journal while Paper 7 awaits NusaRC. Target: Journal of Archaeological Science Reports or similar.

[ ] Song Terus vs Sulawesi gap (~8,000 years): the paper currently attributes this to either survey bias or burial. A third option should be acknowledged: Java may have been first inhabited later than Sulawesi if the crossing route bypassed Java (e.g., via Borneo → Sulawesi directly). However, the Sunda Shelf biogeography makes this unlikely — Java was the most productive and accessible part of the Shelf.

[ ] Toba super-eruption (~74,000 BP): this event predates all current Nusantaran H. sapiens sites. Song Terus (~60,000 BP) postdates Toba. Is there evidence in the Song Terus stratigraphy of a Toba ash layer with occupation above it? This would be direct evidence of survival and recolonization. Check Semah et al. 2023 for stratigraphic detail.

[ ] The 'ghost population' section from v1 (Section 6.2) should be retained in the full paper. It connects the genetic evidence to the taphonomic argument effectively. Not included in this draft for brevity — reinstate in final version.

References

Barker, G., et al., 2007. The 'human revolution' in lowland tropical Southeast Asia: the antiquity of anatomically modern humans at Niah Cave. Journal of Human Evolution 52, 243–261.

Detroit, F., et al., 2004. Palaeontology of Tabon Cave. Comptes Rendus Palevol 3, 705–712.

Gray, R.D., Drummond, A.J., Greenhill, S.J., 2009. Language phylogenies reveal expansion pulses and pauses in Pacific settlement. Science 323, 479–483.

Hansford, J., et al., 2018. Early Holocene human presence in Madagascar evidenced by exploitation of avian megafauna. Science Advances 4, eaat6925.

Hawkins, S., et al., 2017. Late Pleistocene and Holocene human biogeography in island Southeast Asia: ancient DNA from Laili Cave, Timor-Leste. Quaternary Science Reviews 171, 58–72.

Hill, C., et al., 2007. A mitochondrial stratigraphy for island Southeast Asia. Molecular Biology and Evolution 24, 871–882.

Lipson, M., et al., 2014. Reconstructing Austronesian population history in Island Southeast Asia. Nature Communications 5, 4689.

Oktaviana, A.A., et al., 2026. Rock art from at least 67,800 years ago in Sulawesi. Nature 650, 652–656.

Pierron, D., et al., 2017. Genomic landscape of human diversity across Madagascar. PNAS 114, E2341–E2350.

Semah, F., et al., 2023. Song Terus cave and the late Pleistocene record of Java. L'Anthropologie.

Tumonggor, M., et al., 2013. The Indonesian archipelago: an ancient genetic highway linking Asia and the Pacific. European Journal of Human Genetics 21, 824–831.

Westaway, K., et al., 2017. An early modern human presence in Sumatra 73,000–63,000 years ago. Nature 548, 322–325.

Amien, M., 2026a–f. VOLCARCH Series, Papers 1–6. Lab Data Sains, Universitas Bhinneka Nusantara.